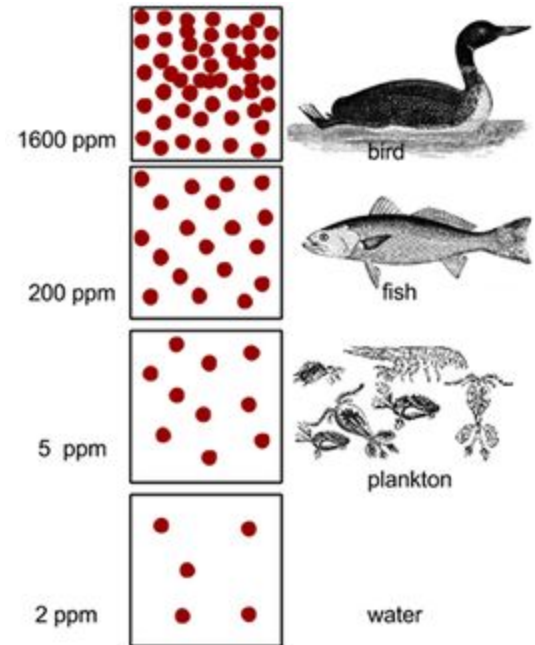


What is Biological Magnification?

Biological magnification refers to the process where toxic substances move up the food chain and become more concentrated at each level. These substances are often pollutants from industries or pesticides from farming. An example of biological magnification and its dangers is any small fish that eats plankton that has been tainted with mercury. Hundreds of small fish might then contain just a few parts of the mercury, not enough to cause major harm. The amount of mercury is measured in ppm, which means “parts per million.”

A bird then might eat hundreds of the small fish, so that now instead of 200 ppm in a single fish, that bird has much higher levels of mercury. The toxin amplifies as it moves up the food chain.

Biological magnification caused a crisis with eagles, where DDT was used to control mosquitoes and other pests. Birds would accumulate toxic levels of DDT in their bodies which would cause their eggs to become fragile and break. The eagle almost became extinct, but lawmakers banned DDT and the eagle is now in recovery.



A. Finding Information: Scan the article for answers to specific questions.

1. What does PPM mean? _____
2. What chemical caused the decline of the eagle population? _____

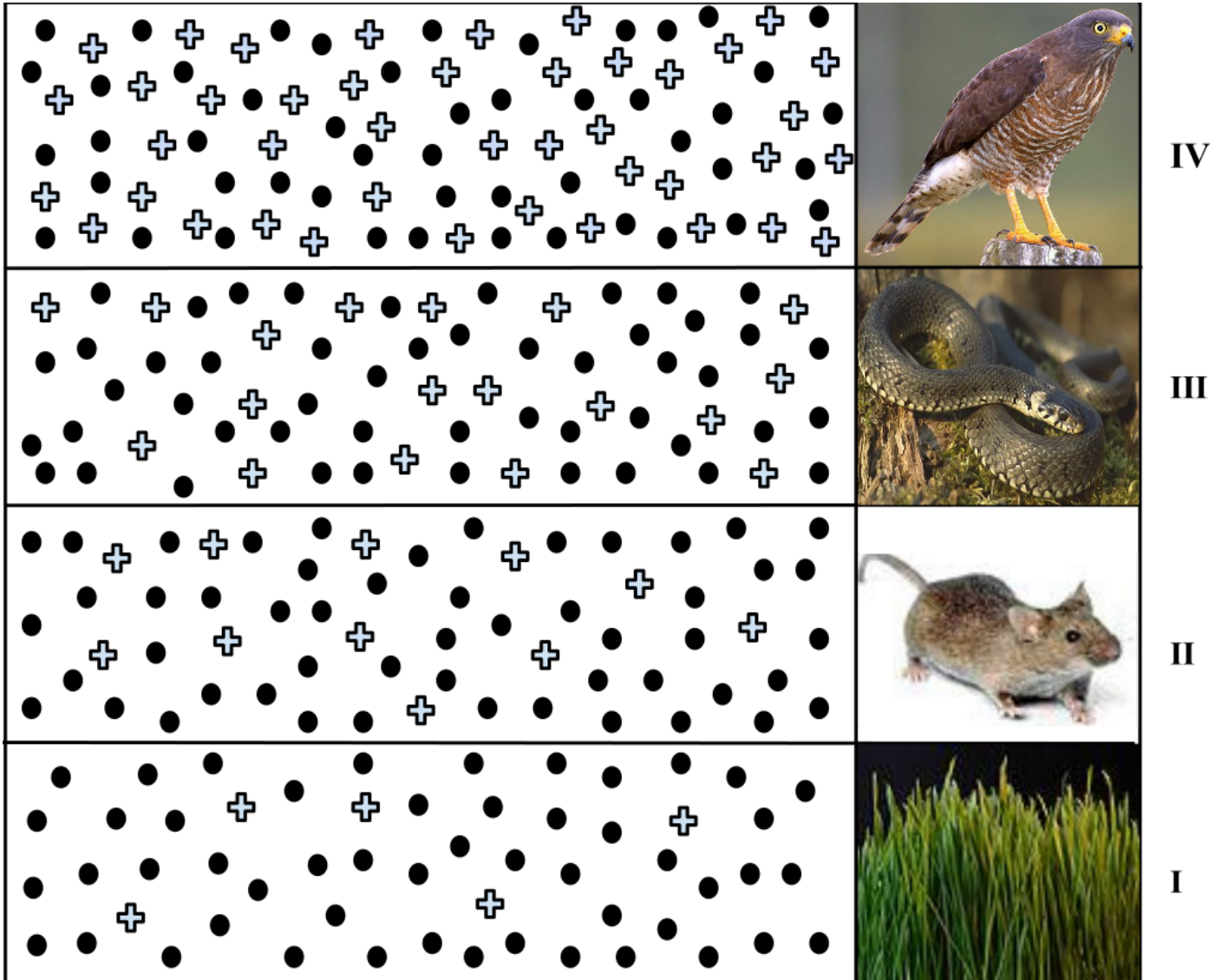
B. Locating evidence and supporting statements. These things are not specifically said.

3. Based on the claims made in the passage, biological magnification is a result of:
 - a) pollution
 - b) climate change
 - c) food shortages
 - d) extinction
4. What saved the eagle population?
 - a) captive breeding
 - b) laws banning DDT
 - c) habitat change
5. What type of animals are most affected by biological magnification?
 - a) herbivores
 - b) omnivores
 - c) aquatic organisms
 - d) top predators
6. What is the author's purpose or point in this writing?
 - a) to convince readers to stop pollution
 - b) to explain the process of biological magnification
 - c) to compare the effects of mercury and DDT
 - d) to describe the effects of mercury poisoning

C. Using Context Clues. Are there words in the passage that you don't understand? Good readers learn to use clues to guess at the meanings of words they don't know.

7. What is plankton?
 - a) a type of toxin
 - b) tiny animals
 - c) a type of bird
8. What does amplify mean?
 - a) to be able to hear
 - b) become sick
 - c) make larger
9. What does accumulate mean?
 - a) to increase
 - b) to chase
 - c) to destroy
10. What does tainted mean?
 - a) colorful
 - b) contaminated
 - c) extinct

D. Application - Using what you have learned to apply to a new situation



11. If the + icons in the image represent a toxic substance. Explain why the hawk is most at risk for being affected by this substance.

12. Pregnant women are advised to limit the amount of fish they eat during pregnancy due to the high amounts of mercury found in some types of fish. Why might some fish have more of this toxin than others?